

2.5 Complex numbers

Further Pure 2 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Write $z=x+iy$, modulus $\sqrt{x^2+y^2}$, argument with quadrant check; polar form $r(\cos \theta + i \sin \theta)$ simplifies multiplication and powers. Conjugation reflects across real axis. Roots of a complex number lie evenly spaced around a circle; De Moivre's theorem gives powers.

Worked example: follow the reasoning

For $z=1+i$, $\text{abs}(z)=\sqrt{2}$, $\arg z=\pi/4$; $z^2=2i$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Find the two square roots of -4 in complex numbers and give their moduli.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

$z^2 = -4$ gives $z = \pm 2i$ [2]; each modulus is 2 [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.